

EAR FALLS, ON, Canada

WMO: 712950

Lat: 50.631N

Lon: 93.221W

Elev: 363

StdP: 97.04

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.5	-29.0	-34.8	0.1	-31.3	-32.2	0.2	-28.7	7.0	-13.4	6.3	-14.9	1.9	270	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	28.4	19.5	26.6	18.4	25.1	17.5	21.3	25.8	20.2	24.2	19.2	23.0	2.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	15.3	23.5	18.9	14.3	22.2	17.7	13.3	21.3	64.0	25.8	60.0	24.4	56.3	22.9	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.7	5.1	DB	-35.5	31.5	2.7	2.0	-37.4	33.0	-39.0	34.2	-40.5	35.3	-42.5	36.8	(4)
(5)			WB	-35.6	23.1	2.6	1.4	-37.5	24.1	-39.0	24.9	-40.5	25.7	-42.4	26.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-16.0	-14.4	-6.1	2.1	9.2	15.4	18.7	17.4	12.3	4.6	-4.1	-13.3	(6)	
	DBStd	13.46	7.80	6.92	7.30	5.56	4.87	3.62	3.26	3.47	4.68	4.73	6.32	7.31	(7)	
	HDD10.0	3657	806	682	500	241	74	3	0	1	29	177	423	722	(8)	
	HDD18.3	5966	1064	916	757	487	283	99	35	59	187	424	673	981	(9)	
	CDD10.0	829	0	0	1	3	51	166	269	229	98	12	0	0	(10)	
	CDD18.3	96	0	0	0	0	2	12	46	30	6	0	0	0	(11)	
	CDH23.3	780	0	0	0	2	23	119	378	218	38	1	0	0	(12)	
	CDH26.7	153	0	0	0	0	2	16	83	47	4	0	0	0	(13)	
(14) Wind	WSAvg	2.2	2.3	2.3	2.4	2.4	2.4	2.1	2.0	2.0	2.1	2.4	2.4	2.2	(14)	
(15) Precipitation	PrecAvg	711	32	26	36	41	62	97	95	102	77	56	43	34	(15)	
	PrecMax	863	86	66	85	112	124	173	199	199	144	156	105	63	(16)	
	PrecMin	484	4	4	2	1	11	36	40	20	20	11	3	12	(17)	
	PrecStd	117	21	14	20	27	29	36	38	47	38	35	24	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	3.2	13.7	19.5	26.1	28.8	31.7	30.5	27.2	21.1	12.4	2.7	(19)	
		MCWB	1.3	1.0	8.3	10.0	14.8	19.3	21.3	20.7	19.0	14.2	9.0	1.4	(20)	
	2%	DB	0.1	0.8	9.1	15.8	23.1	26.2	28.7	27.9	24.0	17.1	8.8	0.3	(21)	
		MCWB	-0.4	-0.7	4.4	8.0	12.9	16.8	19.9	19.8	17.9	12.0	6.0	-0.3	(22)	
	5%	DB	-2.3	-1.1	5.7	13.4	20.7	24.6	27.1	25.8	21.7	14.5	6.2	-1.0	(23)	
		MCWB	-3.0	-2.4	2.4	6.7	12.2	16.3	18.8	18.6	16.5	10.3	4.5	-1.6	(24)	
	10%	DB	-4.8	-4.0	3.2	11.1	18.1	22.6	25.4	23.8	19.6	11.9	4.0	-2.7	(25)	
		MCWB	-5.4	-5.1	0.7	5.5	11.1	15.3	18.0	17.7	15.4	9.3	2.6	-3.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.5	1.3	9.2	11.1	17.3	21.9	23.1	22.9	20.7	15.3	9.5	1.6	(27)	
		MCDB	1.9	2.8	11.7	17.1	21.9	26.3	28.9	26.9	23.9	18.6	11.8	2.4	(28)	
	2%	WB	-0.5	-0.3	5.0	9.1	15.2	19.3	21.5	21.0	19.1	12.7	6.7	-0.1	(29)	
		MCDB	0.1	0.7	8.1	14.7	19.9	23.3	26.4	25.4	22.4	16.0	8.2	0.3	(30)	
	5%	WB	-2.9	-2.4	2.7	7.3	13.7	17.9	20.5	20.0	17.6	11.1	4.4	-1.5	(31)	
		MCDB	-2.4	-1.1	5.5	12.3	18.3	21.4	24.7	23.9	20.7	13.9	5.9	-1.1	(32)	
	10%	WB	-5.4	-5.0	1.0	5.8	12.2	16.7	19.4	18.9	15.9	9.0	2.6	-3.1	(33)	
		MCDB	-4.8	-4.1	2.6	10.7	17.0	20.6	23.2	22.7	18.9	11.4	4.1	-2.7	(34)	
(35) Mean Daily Temperature Range	MDBR		8.7	10.1	10.8	10.8	11.0	10.4	10.5	9.8	8.9	6.8	6.2	7.3	(35)	
	5% DB	MCDBR	8.3	9.6	11.4	14.4	15.3	13.7	12.9	12.4	11.8	11.4	8.4	7.0	(36)	
		MCWBR	7.7	7.9	7.5	8.1	7.2	6.1	5.5	5.7	6.3	6.9	5.9	6.4	(37)	
	5% WB	MCDBR	8.0	8.6	10.8	13.3	11.7	9.9	10.6	10.6	10.1	10.2	7.6	6.7	(38)	
		MCWBR	7.4	7.3	7.5	8.2	6.4	5.7	5.4	5.7	6.4	7.0	5.6	6.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.254	0.260	0.288	0.328	0.346	0.364	0.388	0.362	0.342	0.295	0.276	0.247	(40)	
	taud		2.280	2.288	2.332	2.403	2.455	2.434	2.381	2.464	2.509	2.604	2.460	2.332	(41)	
	Ebn at Noon		788	886	921	916	911	893	863	872	855	849	767	746	(42)	
	Edh at Noon		66	86	100	106	106	109	113	99	85	64	57	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.16	2.38	3.64	5.04	5.32	5.42	5.60	4.68	3.34	1.90	1.01	0.80	(44)	
	RadStd		0.13	0.21	0.31	0.36	0.33	0.44	0.46	0.41	0.35	0.24	0.11	0.10	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page